

Handling Cross-Dialect Syntactic Variation: a Theory-Driven Web Resource

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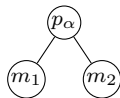
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The Parametric Comparison Method (PCM)

- Quantifies crosslinguistic relations using **abstract syntactic rules** (*parameters*) as *comparanda* (Longobardi & Guardiano 2009)
- Originally conceived for cross-family **(macro-)comparison** (Ceolin et al. 2021)
- Effective for **micro-comparative** analysis (*ultralocality*, Guardiano et al. 2016)

The PCM parameter system

Parameters $\rightarrow$ **abstract** rules encoded in the speaker's grammar
 $\neq$
Manifestations $\rightarrow$ **observable** pattern(s) generated from a rule



Symbolism $\begin{cases} \rightarrow [+]$ the parameter is **active** in the grammar
 $\rightarrow [-]$ the parameter is **absent** from the grammar

Parametric dependencies: one state of parameter α may render the state of parameter β predictable $\rightarrow \beta$ is *neutralised* [0], i.e. contains no information relevant for comparison

Building a web-based resource

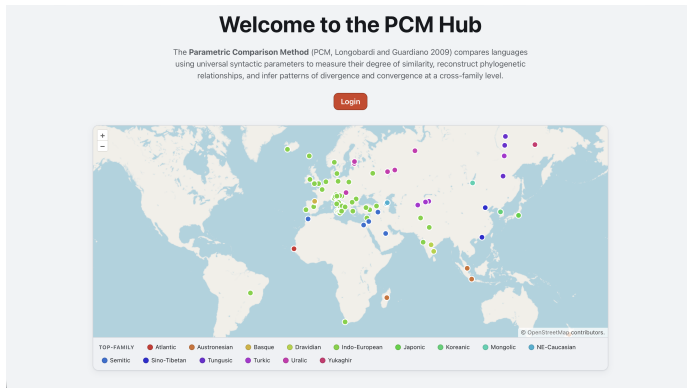
Requirements

- **Intuitive** data recording and sharing
- Structured **network** among data, manifestations and parameters
- Error-free **dependency computation**
- **Expandable** and **modifiable** model
- **Extractable** language and parameter subsets
- Automated computation of language **distances**

Implementations

- **User-oriented** web-based interface
- **Connections** enabling data **checking** and **analysis**
- **Automated** parameter assignment
- **Adaptive architecture**
- Flexible tools for **cross-analysis**
- Built-in **scripts** for taxonomy and phylogeny

The PCM Hub



Access Modalities

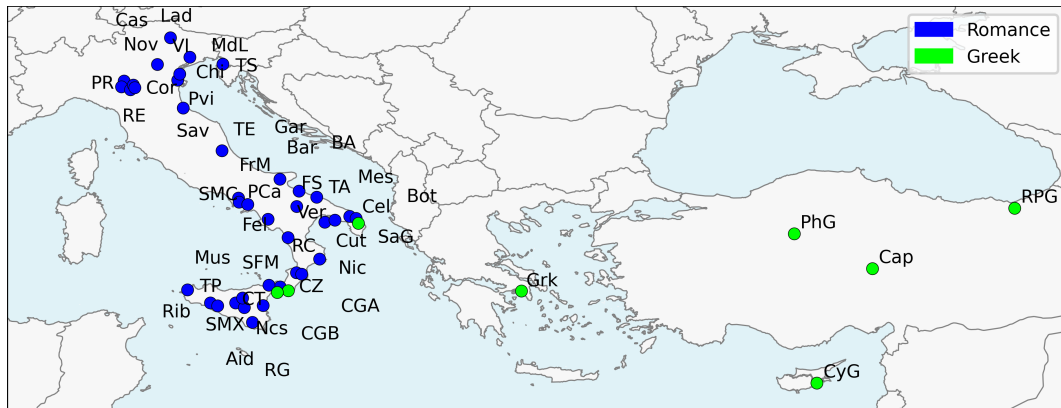
→ Admin

→ User

→ Public



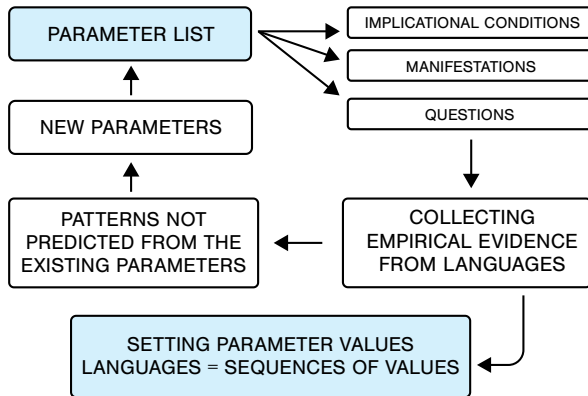
Ultralocality tests: Greek and (Italo-)Romance



Preliminary research questions

- ① Are **inherited patterns** in parameter transmission obscured by horizontal convergence?
 - Does the **distinction between Greek and Romance** emerge from parametric distances?
- ② If yes, does the **internal articulation** of each group align with traditional dialect classifications?

Methodological steps

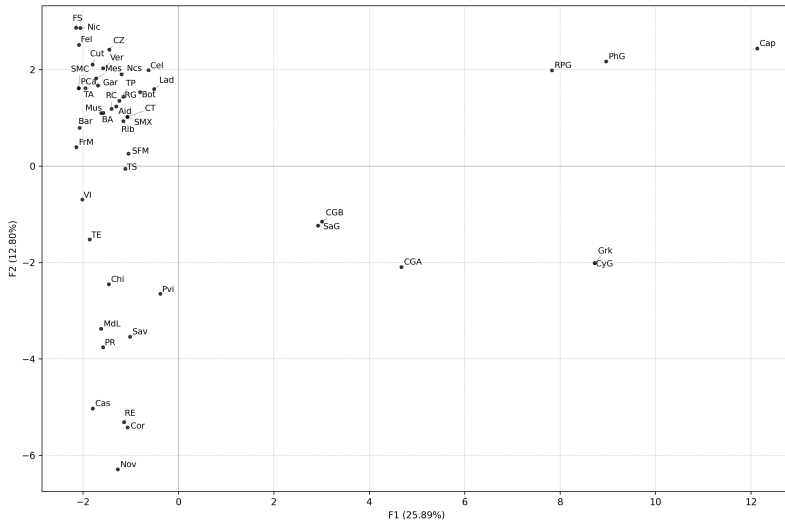


Methodological steps

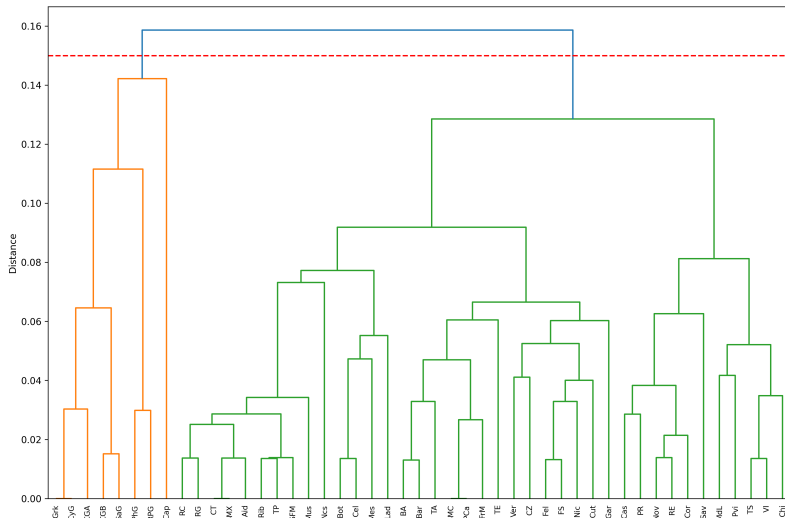
Comparing parameter lists

- Character-based analyses (e.g. PCA)
- Distance-based analyses
 - Parametric distances: $d/(d+i)$ ✓ PCM Hub
 - Evaluating distance distributions (e.g. heatmaps) → Morpheus
 - Cluster analysis (e.g. dendrograms) ✓ PCM Hub
 - Mapping distances against geography ✓ PCM Hub

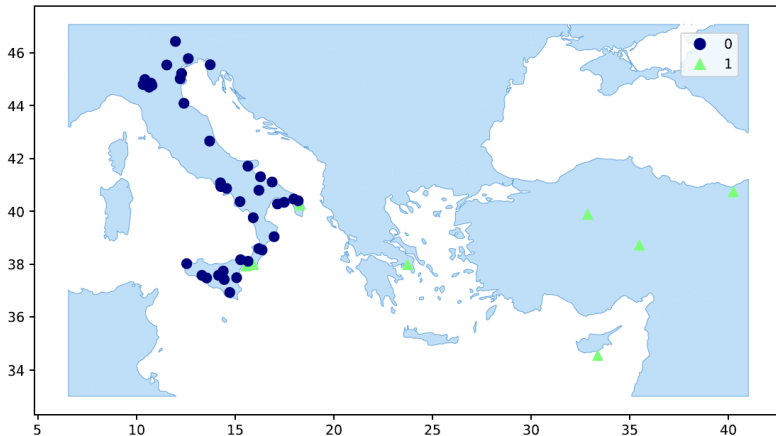
Principal Component Analysis (PCA)



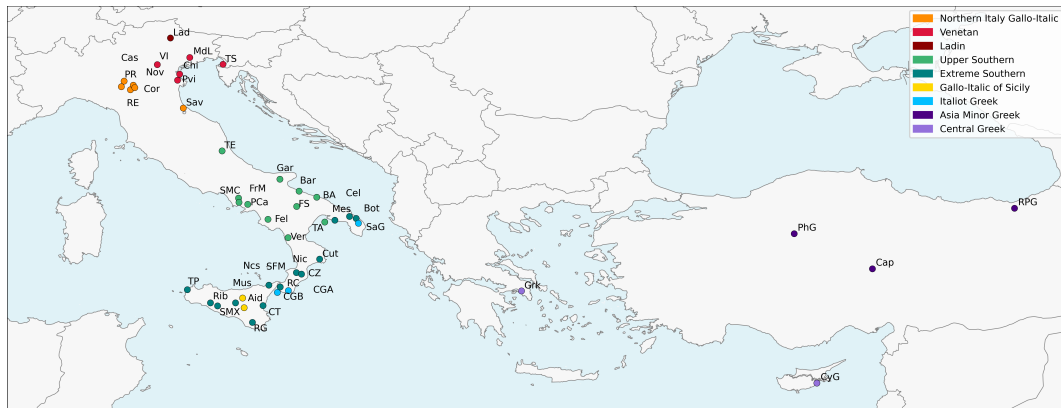
Hierarchical Clustering



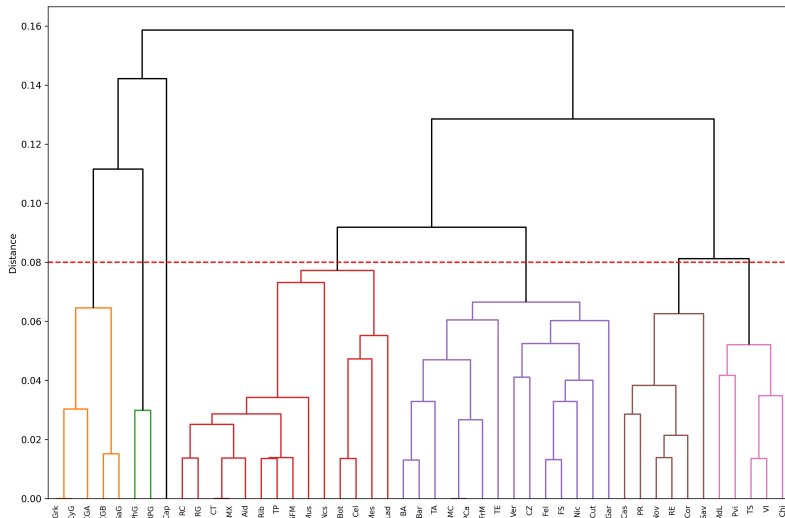
K-means clustering on a map



Monophyletic domains



Monophyletic domains



Thank you!

Contact us at: pcm_lab@unimore.it

You can find scripts and materials at: github.com/GaiaSorge/PCM_Hub

